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Intro to Electrical Engineering

November 7, 2024

Lab Report #7

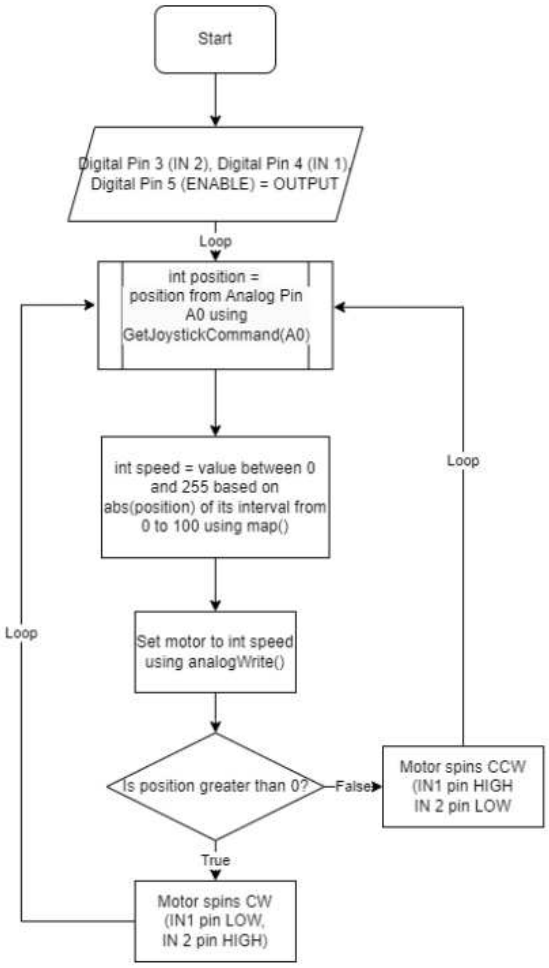
Overview

In this project we will essentially make our joystick control the speed of the fan as well as the direction. The Joystick reads out values from 0 to 1023. We will convert that to -100 to 100. Then convert that so it gives us the voltage values of 0-5. When we toggle the joystick up and down it should start traveling in different directions. The further down or up we go the more the joystick should pick up on its speed.

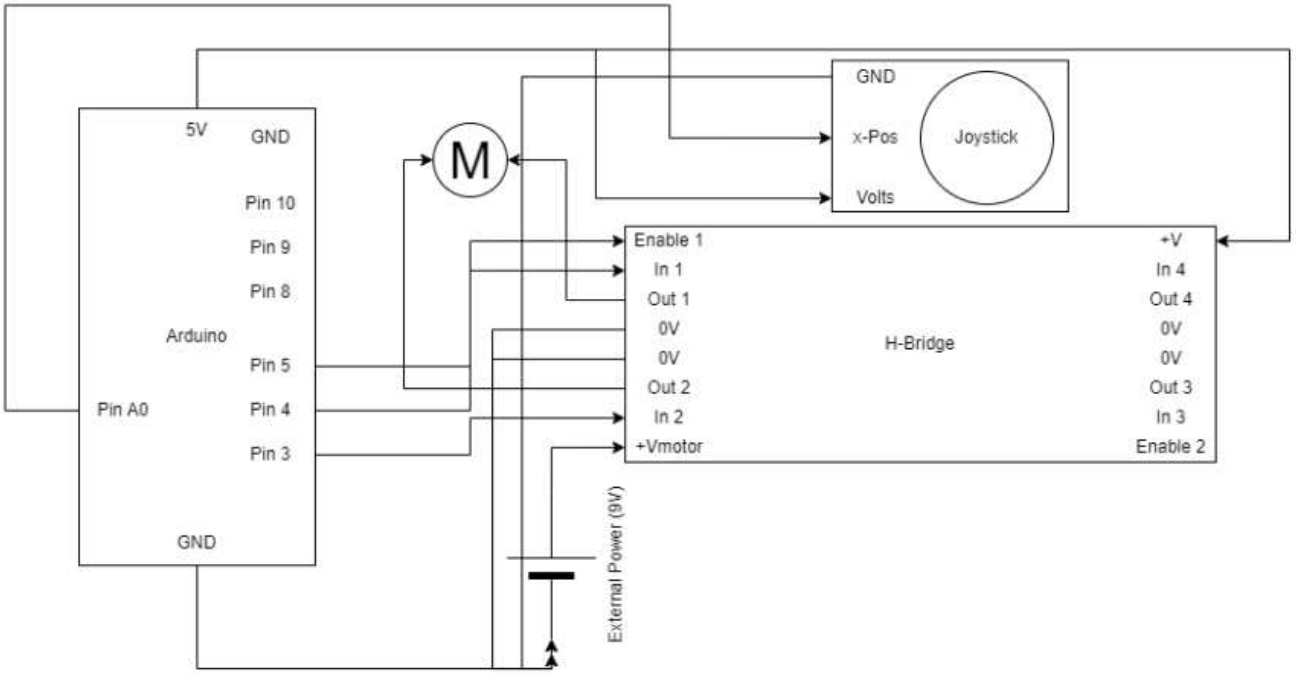
Requirements

I will be using the map commands in my Arduino sketch and this will help me set the proper range with my joystick. The map command will allow me to change the ranges of the values. The joystick will control the value that I initiated as “position”, and if it’s above zero I will enable the current one way while disabling it the other way. If it's below zero I would do vice versa and this will allow me to change the direction of the fan that's moving. The speed will be controlled by the analog read of the pin, and depending how up or down you move the joystick, it will either increase or decrease the speed.

Software Diagram



Wiring Diagram



Design Verifications (Link of working project - https://youtube.com/shorts/XPfCIXu4_BM)

```
int turnON = 5;
int in1 = 4;
int in2 = 3;
int position;
int speed;

void setup() {
  pinMode(turnON, OUTPUT);
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  position = GetJoystickCommand(A0);
  speed = map(abs(position), 0, 100, 0, 255);
  analogWrite(turnON, speed);

  if(position > 0)  {
    digitalWrite(in1, LOW);
    digitalWrite(in2, HIGH);
  } else {
    digitalWrite(in1, HIGH);
    digitalWrite(in2, LOW);
  }
}
int GetJoystickCommand(int pin)  {
  float analog = analogRead(pin);
  float volts = (analog/200) + 0;
  int position = (40*volts)-100;
  delay(1000);
  return position;
}
```

Learnings

I learned a lot more about wiring since that proved to be the main challenging part of this week's lab. It was interesting to see how you could make a circuit and code it in such a way that you can control which way the current is going through. I also got more familiar with code and how to make and initialize variables and then use them effectively in your code. I got more familiar with the map command, as well as making my own variables and plugging them into my code to be able to adjust the speed and direction of the voltage in my project.

Conclusions

One thing that I would do differently is the wiring. I would try to color code the wires next time as at times it can get confusing. I would also try to research more and learn about it, since me and my partner tried to wire for the first time without looking at the given diagram and it proved to be difficult.